

PROJECT TITLE

PRESENTED BY

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OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **Charts used**
- **Insights Drawn**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

- The real estate market generates a large volume of property transaction data, including information about property prices, size, number of bedrooms, location features, and sales timelines. However, this data is often complex and difficult to interpret in its raw form. Real estate investors, brokers, and analysts face challenges in identifying which property characteristics significantly influence property prices and market demand.
- Understanding how factors such as property size, number of bedrooms, waterfront availability, property grade, and seasonal trends affect pricing is essential for making informed investment and pricing decisions. Additionally, identifying patterns across different price ranges and property types is crucial for evaluating market performance and buyer behavior.
- Therefore, there is a need to analyze and interpret real estate transaction data to better understand the key drivers

PROPOSED SOLUTION:

The proposed system aims to analyze real estate transaction data and transform it into meaningful insights that help investors, analysts, and brokers understand market behavior, pricing trends, and profitability drivers. The system uses data analytics and interactive visualization techniques to explore relationships between property characteristics and sales performance. The solution consists of the following components:

1. Data Collection

- The system utilizes historical real estate transaction data containing information such as property price, number of bedrooms, property size, grade, view quality, waterfront availability, and sale dates.
The dataset is obtained from Kaggle and contains detailed property attributes that influence housing prices.

Link : <https://www.kaggle.com/datasets/harlfoxem/housesalesprediction?resource=download>

2. Data Preprocessing

- The collected data is cleaned and prepared before analysis. This includes handling missing values, correcting inconsistencies, and ensuring proper data formatting. Additional calculated fields are created to support analysis, such as price ranges, profit, cost estimates, and return on investment (ROI).

3. Data Modeling and Analysis

- Using Microsoft Power BI, relationships between different variables are explored. Measures and calculations are created using DAX to evaluate key business metrics such as total sales, average price, cost estimation, and profit.

PROPOSED SOLUTION:

4. Data Visualization

- Interactive dashboards are designed to visually represent the analyzed data. Different types of charts and visualizations are used to highlight important patterns and comparisons, including:
- KPI cards for Total Sales, Average Price, Profit, and ROI
- Bar charts for price comparisons by property features
- Scatter plots to analyze the relationship between property size and price
- Line charts to observe monthly price trends
- Donut charts to evaluate waterfront property distribution
- Decomposition Tree to explore key drivers affecting property prices
- These visualizations enable users to interactively explore the data and quickly identify trends and patterns.

5. Dashboard Deployment

- The final dashboard provides a user-friendly interface where users can filter and analyze data using interactive controls such as slicers and drill-down features. This allows stakeholders to examine different property segments, price ranges, and market trends efficiently.

PROPOSED SOLUTION:

- **Result**

- The developed dashboard successfully converts complex real estate transaction data into clear visual insights. Users can easily identify key factors influencing property prices and profitability.
- The analysis reveals that property size, number of bedrooms, property grade, and waterfront availability significantly impact property values. Waterfront homes represent a small portion of properties but contribute to high-value transactions.
- The dashboard also highlights price distribution across different market segments and shows seasonal variations in average property prices. These insights help investors and analysts better understand market dynamics and make more informed real estate decisions.

CHARTS USED:

1. KPI Cards

- **Purpose:**

KPI cards are used to display key performance indicators that summarize the overall performance of the real estate market.

- **Metrics Displayed:**

- Total Sales
- Average Price
- Price per Square Foot
- Assumed Cost
- Profit
- ROI

- **2. Bar Chart – Average Price by Bedrooms**

- **Purpose:**

This chart compares the average price of properties based on the number of bedrooms. It helps understand how property size influences market value.

CHARTS USED:

- **3. Scatter Plot – Price vs Size**
 - **Purpose:**
The scatter plot visualizes the relationship between the property price and the living area size (square feet). It also distinguishes properties based on waterfront availability.
- **4. Line Chart – Average Price by Month**
 - **Purpose:**
This chart displays how the average property price changes across different months, helping identify seasonal patterns in the real estate market.
- **5. Bar Chart – Total Units by Price Range**
 - **Purpose:**
This chart categorizes properties into different price segments and shows the number of units sold in each range.
- **6. Donut Chart – Waterfront Distribution**
 - **Purpose:**
The donut chart illustrates the proportion of waterfront properties compared to non-waterfront properties in the dataset.
- **7. Decomposition Tree**
 - **Purpose:**
The decomposition tree is used to analyze the factors influencing property prices by

INSIGHTS DRAWN:

- The real estate dataset shows **very high market activity with total sales exceeding \$12 billion**, indicating a strong housing market.
- **Average house prices increase as the number of bedrooms increases**, but the growth slows after 7-8 bedrooms, suggesting diminishing returns for very large homes.
- There is a **positive relationship between property size and price**, meaning larger houses generally sell for higher prices.
- **Waterfront properties are very rare**, making up less than 1% of the total properties, but they command significantly higher prices.
- The majority of property transactions occur in the **\$200K-\$600K price range**, indicating that mid-range homes dominate the market.
- Monthly price trends show **moderate fluctuations throughout the year**, which may indicate seasonal buying behavior.
- Property **grade and view quality strongly influence price**, with higher-grade houses and better views leading to significantly higher property values.

CONCLUSION:

- The Real Estate Sales Dashboard developed using Microsoft Power BI provides a comprehensive analysis of property transaction data and highlights important trends within the housing market. By transforming raw real estate data into interactive visualizations, the dashboard enables users to easily understand key factors influencing property prices and market performance.
- The analysis reveals that property characteristics such as **number of bedrooms, house size, grade, view quality, and waterfront availability** significantly impact property values. Mid-range properties dominate the market in terms of transaction volume, while luxury properties contribute higher individual sale values. Seasonal variations in prices also indicate changing buyer demand throughout the year.
- Overall, the dashboard helps investors, brokers, and analysts gain meaningful insights into real estate market behavior. It supports better decision-making by providing clear visual representations of pricing trends, profitability indicators, and property value drivers.

FUTURE SCOPE:

- The current dashboard provides valuable insights into real estate transaction patterns; however, it can be further enhanced with advanced analytical features and additional data sources.
- **Predictive Price Modeling**
Machine learning models could be integrated to predict future property prices based on historical trends.
- **Geographical Analysis**
Adding map-based visualizations would help analyze property values across different regions and neighborhoods.
- **Real-Time Data Integration**
Connecting the dashboard to live real estate data sources would allow real-time market monitoring.
- **Advanced Investment Analytics**
Additional metrics such as rental yield, property appreciation rate, and risk indicators could support deeper investment analysis.
- **User Personalization**
Future versions could include customized filters and user-specific dashboards for investors, brokers, and analysts.

REFERENCES:

Kaggle link :

<https://www.kaggle.com/datasets/harlfoxem/housesalesprediction?resource=download>

GitHub link:

https://github.com/sam05maurya/Real_Estate_Sales_Dashboard

Dashboard link:

https://github.com/sam05maurya/Real_Estate_Sales_Dashboard/blob/main/RealEstate_Sales_dashoard.pbix

Thank You